CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 13: Section and Mid-Point Formula

Every formula and result of the chapter on one sheet

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|---|---|---|
| 1 | Section formula $(m_1 : m_2)$ | $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$ |
| 2 | Section formula $(k : 1)$ | $\left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1}\right)$ |
| 3 | Mid-point formula | $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ |
| 4 | Centroid of a triangle | $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$ |
| 5 | Centroid on a median | Divides it in the ratio $2 : 1$ from the vertex |
| 6 | Points of trisection | Ratios $1 : 2$ and $2 : 1$ |
| 7 | Point on the x-axis | Ordinate $= 0$ |
| 8 | Point on the y-axis | Abscissa $= 0$ |
| 9 | Parallelogram | The diagonals have the same mid-point |
| 10 | Centre of a circle | Mid-point of any diameter |